AEIOUFramework	Group ID: E-23	Sheet No: 1
Activity Sheet	Group Name: MindCraft	
	Project Title: “Quietify”	

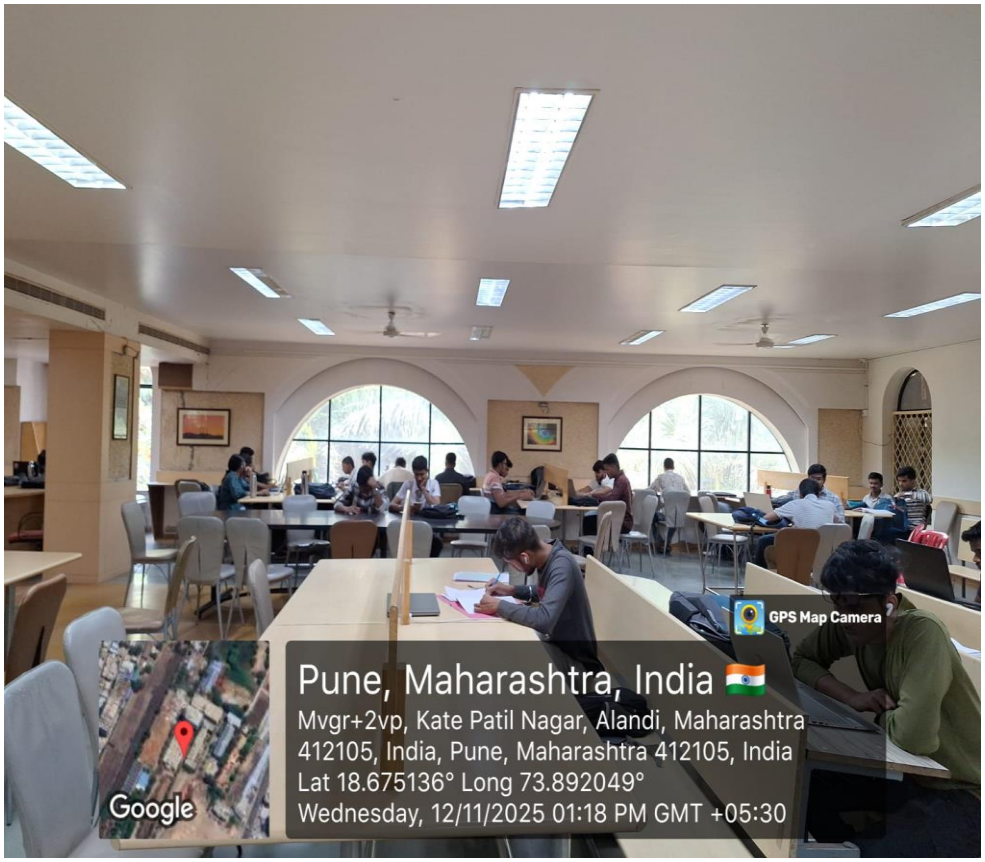
Things Happening Around Product Concept:

- Students create noise by talking or moving chairs.
- Noise disturbs others in the library.
- Librarians cannot monitor all areas continuously.
- Microphone sensor detects noise levels.
- Alert activates when noise crosses the limit.
- message reminds users to reduce noise.
- Helps maintain a quiet library environment.

Sketch/Photo- Summary of Activities:

Effect of Activity on Product, Special Notes:

- Environmental sound affects concentration in the library.
- User behavior helps identify acceptable noise levels.
- Ideal noise threshold is around **60 dB**.
- System promotes self-discipline among users.
- Reduces need for librarian intervention.
- Improves overall user experience.
- Maintains discipline in shared spaces.



AEIOU Framework	Group ID: E-23	Sheet No: 2
Environment Sheet	Group Name: MindCraft	
	Project Title: “ Quietify”	

Season, Time, Personal, Location For Product

Season effect:

Rainy: Slight increase in background noise due to rain, system still detects human noise accurately.

Summer: Library is crowded during exams; noise levels increase, system helps maintain silence..

Winter: Fewer students; noise levels are low, system shows more “OK” readings.

Scenes /Photos of Environment Related to Concept:

Autumn: No Effect
Spring: No Effect

Time Effect:-

Day time : Library is full, conversations and chair movements increase — system will detect frequent noise and display
alerts.

Morning: Quiet hours - system stays in “OK” mode most of the time.

Afternoon: Moderate activity — system will alert occasionally.

Evening: More students visit; slightly higher noise detected.

Night: If library open, it’s calmer — system detects very few alerts.

Late night: Mostly silent; device confirms quiet atmosphere.

Wee hrs/ Early morning: Very peaceful — system remains inactive unless cleaning staff or movement occurs.

Personal effect:-

Kid: Kids may get curious seeing the display or LED light up. The system can teach them about maintaining silence in study areas.

Teenager: Teenagers are usually more active and tend to talk in groups. The system acts as a reminder to lower their voices and maintain discipline.

Adult: Adults (college students, staff) will understand the purpose of the alert and adjust their behavior accordingly. Helps them focus better while studying or reading.

Senior citizen: Appreciate quieter environment; enjoy peaceful reading.

Male: May ignore small noises visual alert increases awareness.

Female: More sensitive to noise value calm and silent surroundings.

Location effect:

- Device works best in quiet indoor spaces.
- Suitable for libraries, study halls, and reading rooms.
- External noise near entrances may cause false alerts.
- Central quiet zones are ideal for installation.
- Can also be used in offices, labs, and self-study rooms.

AEIOU Framework	Group ID: E-23	Sheet No: 3
Interaction Sheet	Group Name: MindCraft	
	Project Title: “ Quietify”	

Users Expectation About Product:

- The product should accurately detect when the noise level in the library becomes too high.
- It should alert users in a simple and non-disturbing way (like a light or display message).
- The system must not make additional noise while warning others.
- It should be compact, easy to install, and visible to everyone.

- The device should clearly display when the library environment is “Quiet” or “Noisy.”

- It should work automatically without requiring constant human supervision.

- Users expect the system to help maintain discipline and improve focus inside the library.

Summary of Interaction:

•Users find it difficult to maintain silence during peak study hours.
•Many users talk without realizing their noise level.
•Visual alerts like **“NOISE”** help users become aware instantly.

- Librarians appreciated reduced need for manual monitoring.
- System should be accurate, simple, and visually noticeable.
- Feedback helps refine noise thresholds for library use.
- Students responded positively to the concept.

AEI**O**U Framework	Group ID: E-23	Sheet No: 4
Object Sheet	Group Name: MindCraft	
	Project Title: “ Quietify”	

List of Components:

1. Arduino Board (Nano).
2. 9V DC Power Supply.
3. Noise Detector Sensor (sound sensor)
4. Low Noise Indicator (Green LED)
5. Moderate Noise Indicator (Yellow LED)
6. High Noise Indicator (Red LED)
7. ON–OFF Switch.
8. Display.

List of Inventory with Photos:

Arduino Board (Nano)

9V DC Power Supply

Noise Detector Sensor (sound sensor)

Low Noise Indicator (Green LED)

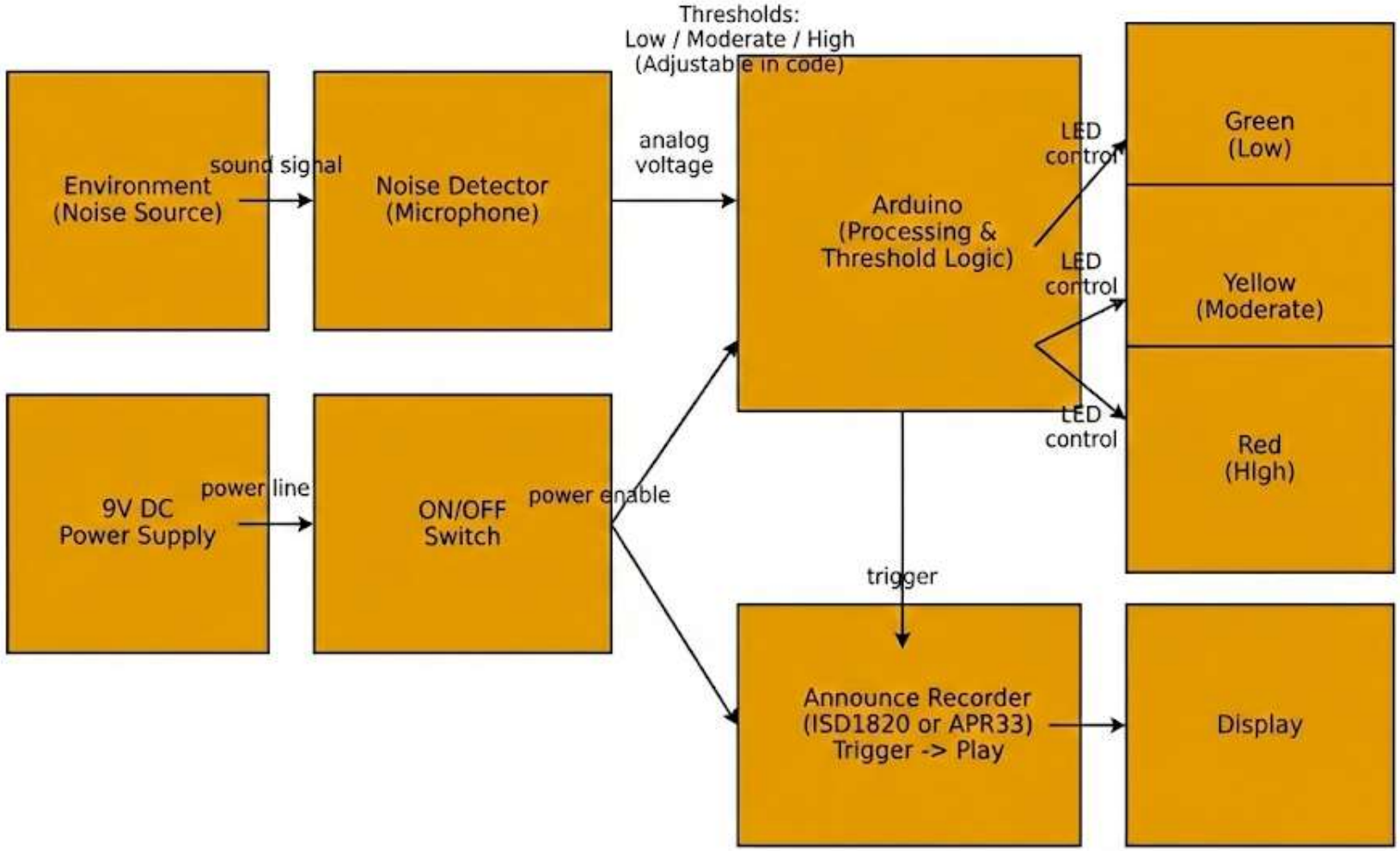
Moderate Noise Indicator (Yellow LED)

High Noise Indicator (Red LED)

ON–OFF Switch

Display

Diagram Showing Interaction Between Objects :



AEIO**U** Framework	Group ID: E-23	Sheet No: 5
User Sheet	Group Name: MindCraft	
	Project Title: “ Quietify”	

Identified Potential Users of Product:

Teachers / Librarians*:*

•Monitor library noise without constant supervision.

•Reduce need to scold students manually.

Student:

•Benefit from a quieter study environment.

•“NOISE” alert helps self-regulate voice levels.

Hotel Manager:

•Can use system in conference rooms or quiet lounges.

Existing User Photographs / Special Notes:

Branch Manager:

In corporate offices, it can be used to ensure a calm work environment or during meetings.

Officer Managers:

Helps maintain discipline and silence in workplaces such as administrative offices or training halls.

Scientists:

Can use this type of system in research labs where silence is necessary for concentration and sensitive instruments.

Kids:

When installed in school libraries or classrooms, it can teach children the importance of silence and discipline in
shared spaces.

Special Modification For Users/ Method of Use:

- Product can be modified for different users and environments.
- In libraries, only visual alerts (LED/display) are used to maintain silence.
- In offices or hotels, system can connect to a central monitor.
- Device is installed at the center of the room.
- Powered through USB.
- Noise threshold is set as required.
- System works automatically with no manual operation.

Empathy Mapping

Group ID:

Sheet No: 6

Group Name: Mindcraft

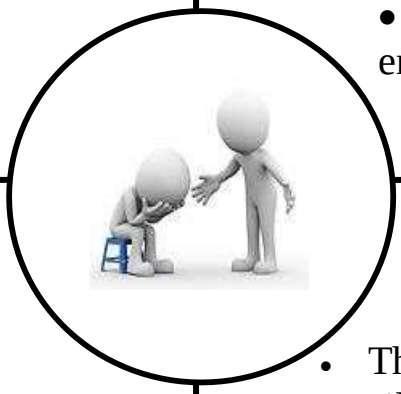
Project Title:“ Quietify”

Hear

- Other students talking loudly
- Phone notification sounds
- Chairs dragging
- Whispering turning into chatter
- Group discussion noise

Users

- Students studying in the library
- Librarians who want to maintain silence
- Exam-preparing students
- Readers who need quiet environment



Think & Do

- Think: “I can’t focus here”, “Why can’t people stay quiet?”
- Think: “I wish there was a system to control noise.”
- Do: Move to another corner.
- Do: Use earphones even though not allowed sometimes.
- Do: Tell others to be quiet.

See & Feel

- They see people not following silence rules.
- They feel irritated and distracted.
- They feel stressed before exams.
- They see librarians repeatedly asking for silence.
- They feel helpless because they can’t control others.

Sad Story:

“A student was preparing for an exam but constant talking and phone sounds disturbed him. He had to change his seat multiple times and still couldn’t focus.”

Happy Story:

“After installing the Library Noise Detection System, students study peacefully. Whenever noise crosses a limit, the indicator alerts everyone, and people automatically lower their voice.”

POV Sheet

USER	***NEED***	***INSIGHT***
Students	A quiet place to study without disturbances	Students lose focus even with small noises, so an automatic alert system can help maintain silence better than manual warnings.
Librarian	A simple way to monitor and control noise levels	Librarians cannot supervise the whole library all the time; a noise detection system helps them maintain discipline easily.
Exam-preparing students	A peaceful, distraction-free environment	During exams, students become more sensitive to noise, so real-time noise alerts help them stay concentrated.
College/Institution	A system that ensures strict silence rules	Institutions need an automated way to promote discipline and improve the study environment without extra staff effort.

Ideation Canvas	Group ID: E-23	Sheet No: 7
Group Name: MindCraft.	Project Title: "Quietify"	

People

Students, Teachers and Librarian

Activities

- Reading and self-study
- Group discussions / doubt clearing
- Moving chairs and books
- Entering and exiting library
- Using phones or laptops

Situation/Context/Location

Situation:

- Library during exam hours
- Crowded study halls
- Quiet reading zones
- Rainy days (background noise)
- Peak study time vs normal hours

Props/Possible Solutions

- Microphone-based noise sensor device
- LED display showing **“QUIET” / “NOISE”**
- Color indicator (Green = quiet, Red = noisy)
- Adjustable noise threshold (around 60 dB)
- Automatic alert system (no human control needed)
- Wall-mounted or ceiling-mounted device
- USB powered / low power consumption
- Central monitoring option (for librarians)

Product Development Canvas	Group ID: E-23	Sheet No:8
Group Name: MindCraft	Project Title: “ Quietify”	

Purpose	Components	Customer Revalidation
• To maintain silence in libraries • To reduce noise disturbance for students • To help students concentrate better • To support librarians in controlling noise • To create a disciplined study environment	• Sound sensor • Microcontroller (Arduino nano) • Buzzer • LED indicator • Power supply • Connecting wires and breadboard	• Students report better concentration • Librarians find noise control easier • Reduced complaints about noise • Positive response from management

People

- Students studying in the library
- Exam-preparing students
- Librarians
- College/School management
- Visitors using the library

Product Features

- Detects noise level in real time
- Alerts when noise exceeds limit
- Visual indication using LEDs
- Audio alert if required
- Simple and easy to use
- Low power consumption

Product Experience

- Students feel focused and calm
- Less distraction during study
- Librarians feel relaxed and supported
- Automatic silence enforcement
- Peaceful and disciplined environment

Reject, Redesign, Retain

- **Retain:** Noise detection feature
- **Retain:** LED visual alert
- **Redesign:** Alert volume level
- **Redesign:** Sensor sensitivity
- **Reject:** Continuous loud buzzer